Ocean Literacy at Journey School

A pilot program for school-wide marine science education

Effi Bar · Marine Biologist & Educator

Why we're here

- Journey's mission speaks of *connection to the natural world*
- Guanacaste students live near the ocean — but rarely engage with it scientifically
- Today: I want to **listen first**, then share what a school-embedded program could look like

> *"Before I show you anything — how do you currently approach marine and environmental education? What's working? What feels missing?"*

Why this fits Journey specifically

Three reasons — any one would be enough

- **Standards-aligned, inside the existing curriculum.** Built on the Ocean Literacy framework (NOAA / NMEA), mapped to **IB PYP** *Sharing the Planet* and **MEP** biodiversity standards. We integrate — we don't ask for a new slot.
- **A real draw for parents.** A school-embedded Ocean Literacy program is a visible differentiator for admissions, with a credential pathway (FEE Green Flag → MSA / Cognia portfolios).
- **Career exposure starting middle school.** Most students don't know marine careers exist — boat captain, researcher, conservation officer, marine policy. The program lets them *picture themselves* in those futures.

What is Ocean Literacy?

An internationally recognized framework — not my invention.

- Developed by **NOAA / NMEA**, used in 100+ countries
- **7 core principles**, scaffolded across K–12 with 45 supporting concepts
- Maps directly onto **IB PYP** *Sharing the Planet* and **MEP** *Ciencias Naturales — biodiversidad marina*

> *"MEP mandates that biodiversidad marina be taught — but most schools fulfill it with terrestrial content because they don't have a marine biologist. I fill that gap."*

The 7 Principles of Ocean Literacy

1. Earth has one big ocean with many features
2. The ocean shapes the features of Earth
3. The ocean drives weather and climate
4. The ocean made Earth habitable
5. The ocean supports a great diversity of life
6. The ocean and humans are inextricably interconnected
7. The ocean is largely unexplored

How it scales by grade

Grade	Frame	What students *do*
K–2 (PYP early)	*The ocean is alive, and I've met it*	Storytelling · touch tanks · habitat models · tide-pool encounters
3–5 (PYP late)	*The ocean is a system, and I can describe it*	Ecosystems · beach-debris counts · water-clarity logs · vocabulary build
6–8 (MYP)	*The ocean has problems, and I can investigate them*	Water-quality testing · citizen science · ecosystem dynamics
9–12 (DP / CAS)	*The ocean has answers, and I can find them*	Research-question fieldwork · marine policy · NOAA / PRETOMA datasets

> *"The pilot enters at one tier — wherever you have the strongest energy. The framework scales to the rest in Year 2."*

What students actually do

- **Hands-on, 90 min lessons** — rocky-shore models, specimen observation, plankton towing
- **Field excursions** — Avellanas/Negra tide pools · sea turtle nesting · mangroves · Gulf of Nicoya
- **Real data collection** — water clarity · beach debris · microplastics · citizen science contributions
- **Worksheets, structured experiments, class discussions** — every session has a teacher guide

The local advantage

We have a world-class natural laboratory 20 minutes away

- **Olive ridley arribadas** — Aug–Nov, Ostional. *45 minutes from Tamarindo.*
- **Leatherbacks** (Critically Endangered) — Oct–Mar, Las Baulas. *15 minutes from Tamarindo.*
- **Both happen during the school year.**

> *"No PowerPoint beats — 'the turtles are nesting 20 minutes from here, and your Grade 7 class can document it.'"*

Two ways in

Option 1 — Full year (recommended)

- Begins December · 9–10 months
- Monthly visits with marine biologist
- 9 × 90-min hands-on lessons
- 3 field excursions (one per term)
- 2–3 teacher PD sessions (train-the-trainer)
- Worksheets, field guides, differentiation materials

Option 2 — Single project

- One ocean literacy project per grade level · 1 field excursion
- Project-based assessments · 2 teacher training sessions
- Number of lessons adapted to scope

About me

Effi (Efrat Bar) — marine biologist & environmental educator · 10+ years

- **B.Sc. Marine Sciences** (with honors) · graduate work in marine protected area management
- **EcoOcean** — ecological surveys, biodiversity documentation, marine reserve planning
- **Educational programs** — classroom & field, including aboard research vessels
- **Public outreach** — sea turtle rescue & rehabilitation center

> *"I live here. My family is rooted here. The program is designed to be renewable — and I build teacher capacity so the school keeps the framework regardless of me."*

What I'd love to leave with today

1. The **one problem** in marine/environmental education you can't solve with what you have today
2. The **age band(s)** you'd prioritize for the pilot
3. The **shape** that fits your calendar — weekly · special days · camps · transversal
4. Whether you'd like a **tailored written proposal this week**

> *"I want it to be something built for Journey specifically — not a template. That's why I'm asking these four questions before I write it, not after."*

Thank you

Questions?

Effi Bar · kite35@gmail.com